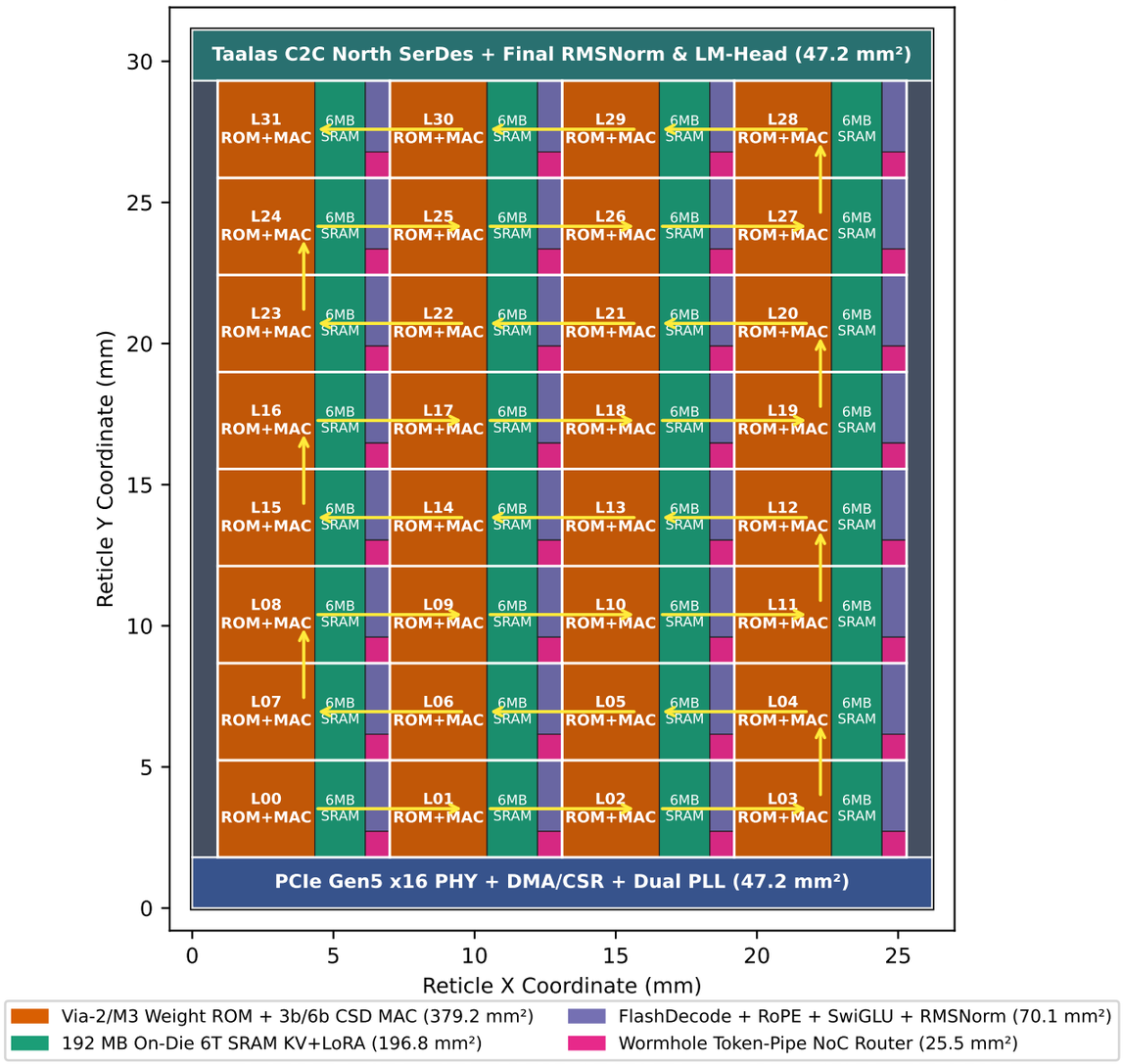
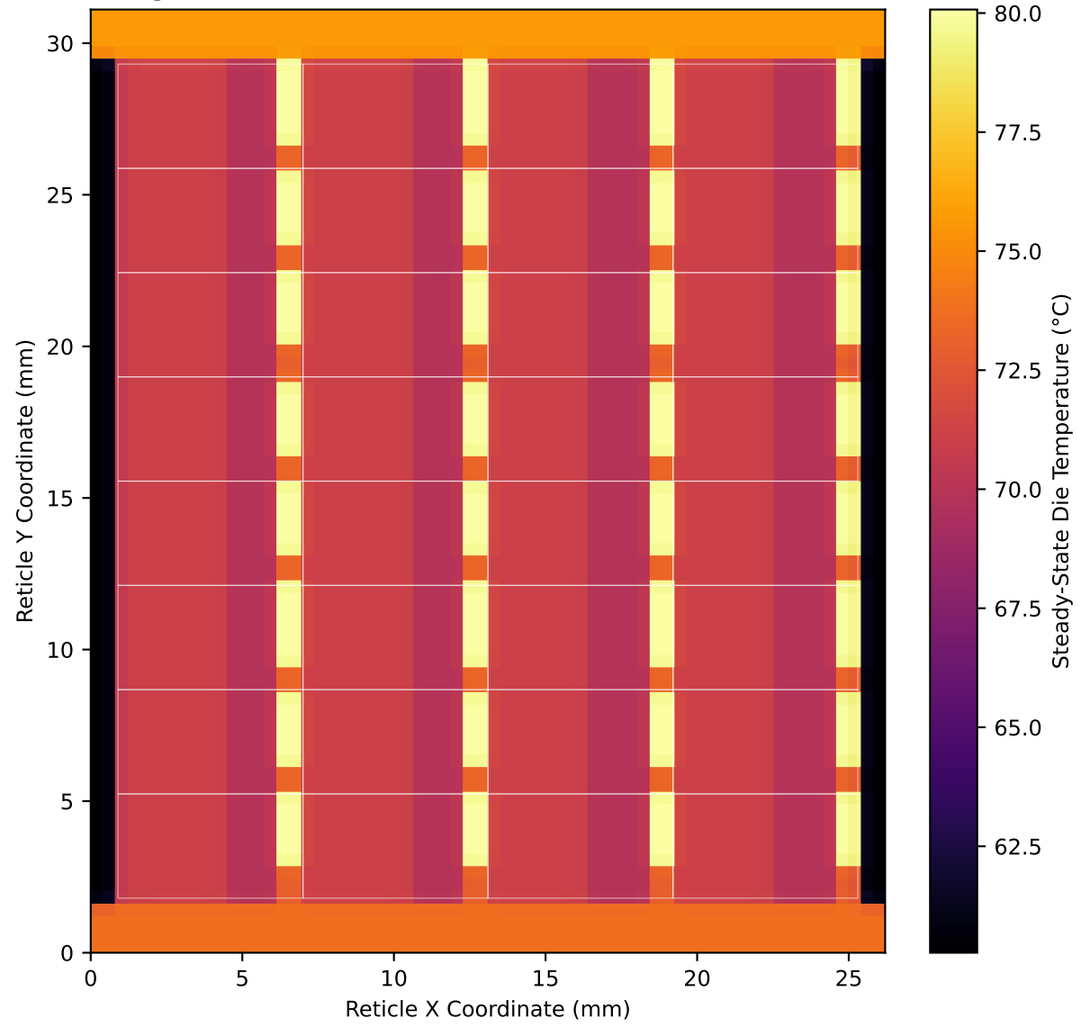
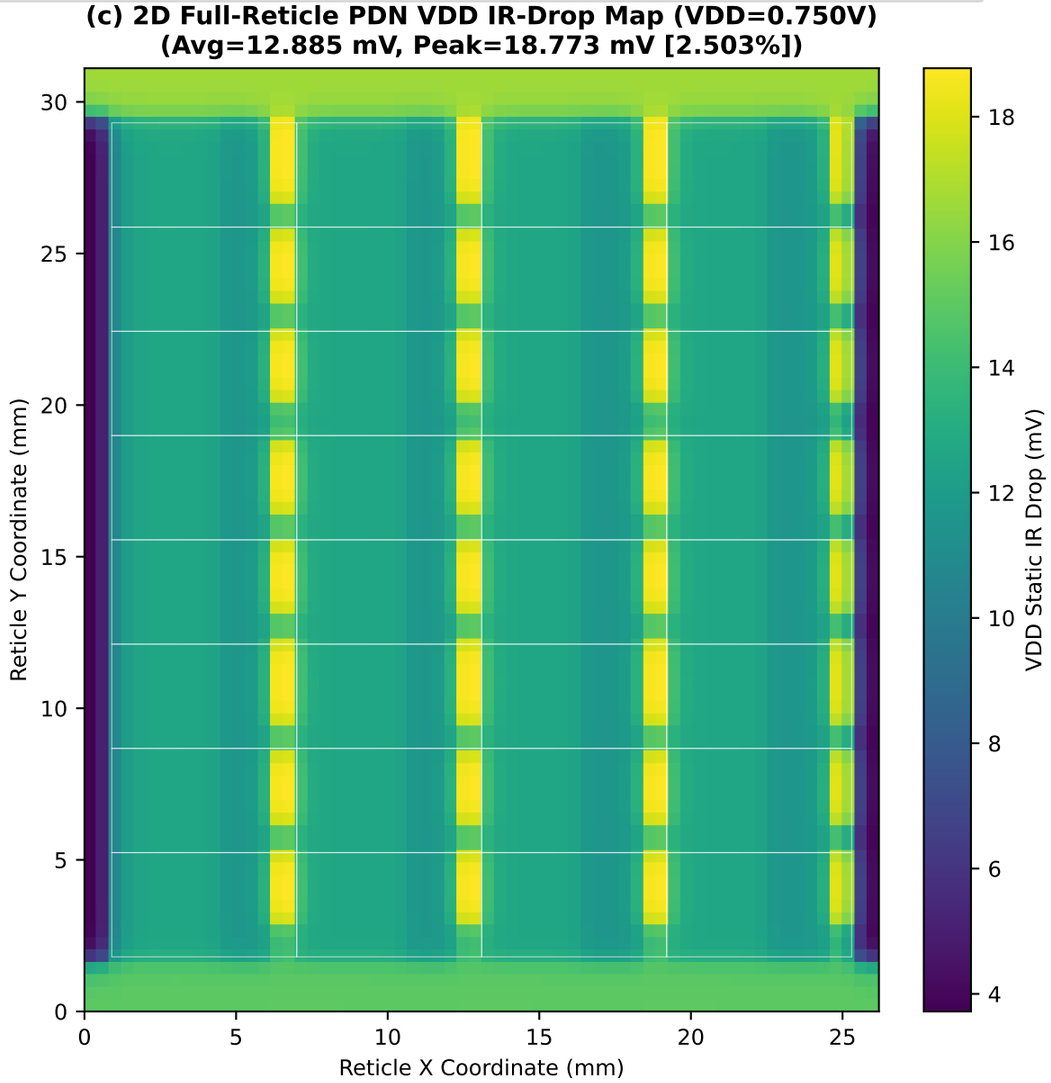
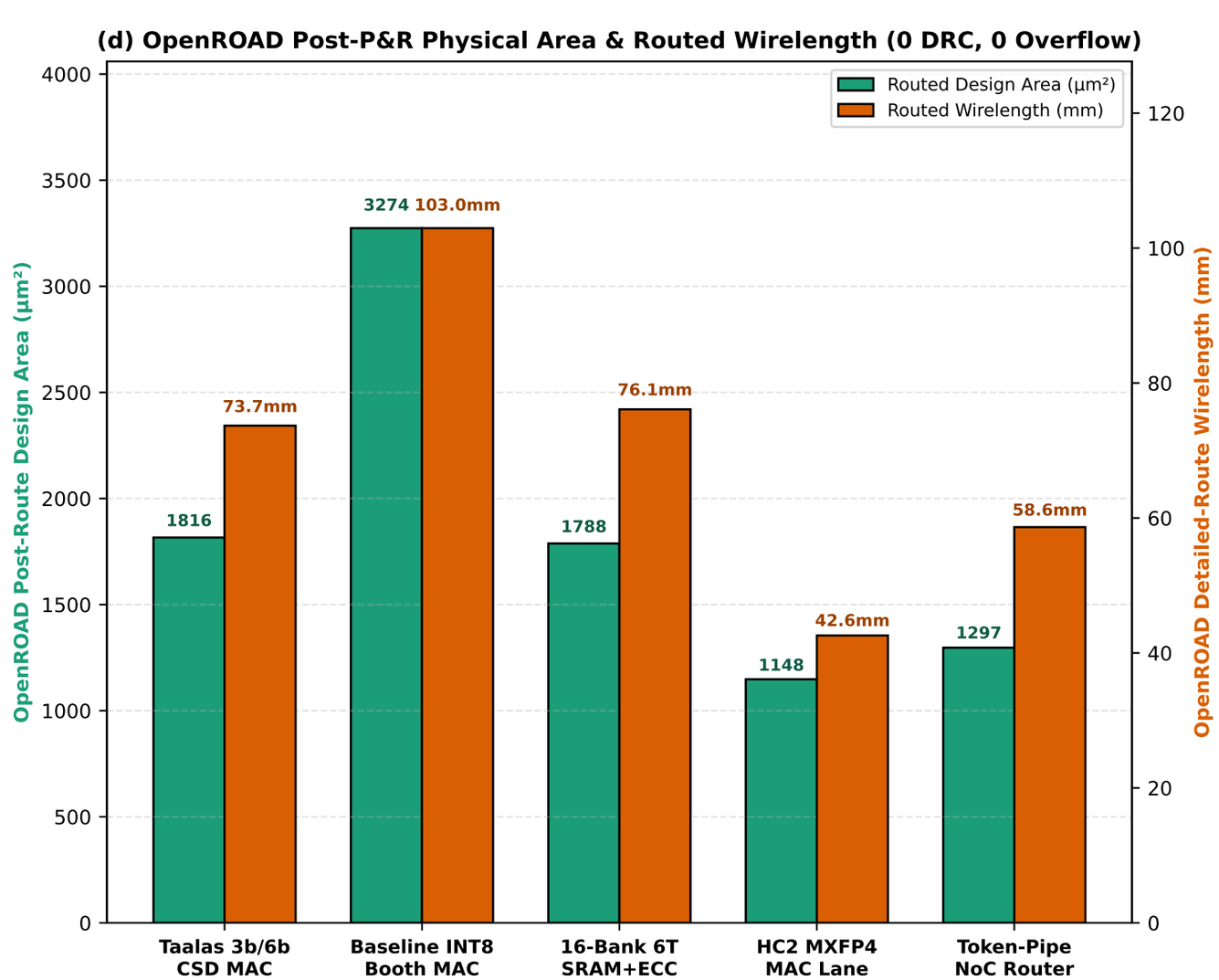

(a) Taalas HC1 815.0 mm² (26.20×31.11 mm) 32-Layer Reticle Floorplan
Taalas C2C North SerDes + Final RMSNorm & LM-Head (47.2 mm²)
PCIe Gen5 x16 PHY + DMA/CSR + Dual PLL (47.2 mm²)
L31 ROM+MAC
L30 ROM+MAC
L29 ROM+MAC
L28 ROM+MAC
L24 ROM+MAC
L25 ROM+MAC
L26 ROM+MAC
L27 ROM+MAC
L23 ROM+MAC
L22 ROM+MAC
L21 ROM+MAC
L20 ROM+MAC
L16 ROM+MAC
L17 ROM+MAC
L18 ROM+MAC
L19 ROM+MAC
L15 ROM+MAC
L14 ROM+MAC
L13 ROM+MAC
L12 ROM+MAC
L08 ROM+MAC
L09 ROM+MAC
L10 ROM+MAC
L11 ROM+MAC
L07 ROM+MAC
L06 ROM+MAC
L05 ROM+MAC
L04 ROM+MAC
L00 ROM+MAC
L01 ROM+MAC
L02 ROM+MAC
L03 ROM+MAC
6MB SRAM
Reticle Y Coordinate (mm)
Reticle X Coordinate (mm)
Via-2/M3 Weight ROM + 3b/6b CSD MAC (379.2 mm²)
FlashDecode + RoPE + SwiGLU + RMSNorm (70.1 mm²)
192 MB On-Die 6T SRAM KV+LoRA (196.8 mm²)
Wormhole Token-Pipe NoC Router (25.5 mm²)


(b) 2D Steady-State Thermal Map @ 218.8W Nominal TDP
(Avg=71.36°C, Peak=80.07°C < 85°C Air-Cooled Limit)
Reticle Y Coordinate (mm)
Reticle X Coordinate (mm)
Steady-State Die Temperature (°C)


(c) 2D Full-Reticle PDN VDD IR-Drop Map (VDD=0.750V)
(Avg=12.885 mV, Peak=18.773 mV [2.503%])
Reticle Y Coordinate (mm)
Reticle X Coordinate (mm)
VDD Static IR Drop (mV)


(d) OpenROAD Post-P&R Physical Area & Routed Wirelength (0 DRC, 0 Overflow)
Routed Design Area (µm²)
Routed Wirelength (mm)
OpenROAD Post-Route Design Area (µm²)
OpenROAD Detailed-Route Wirelength (mm)
1816
73.7mm
3274
103.0mm
1788
76.1mm
1148
42.6mm
1297
58.6mm
Taalas 3b/6b CSD MAC
Baseline INT8 Booth MAC
16-Bank 6T SRAM+ECC
HC2 MXFP4 MAC Lane
Token-Pipe NoC Router